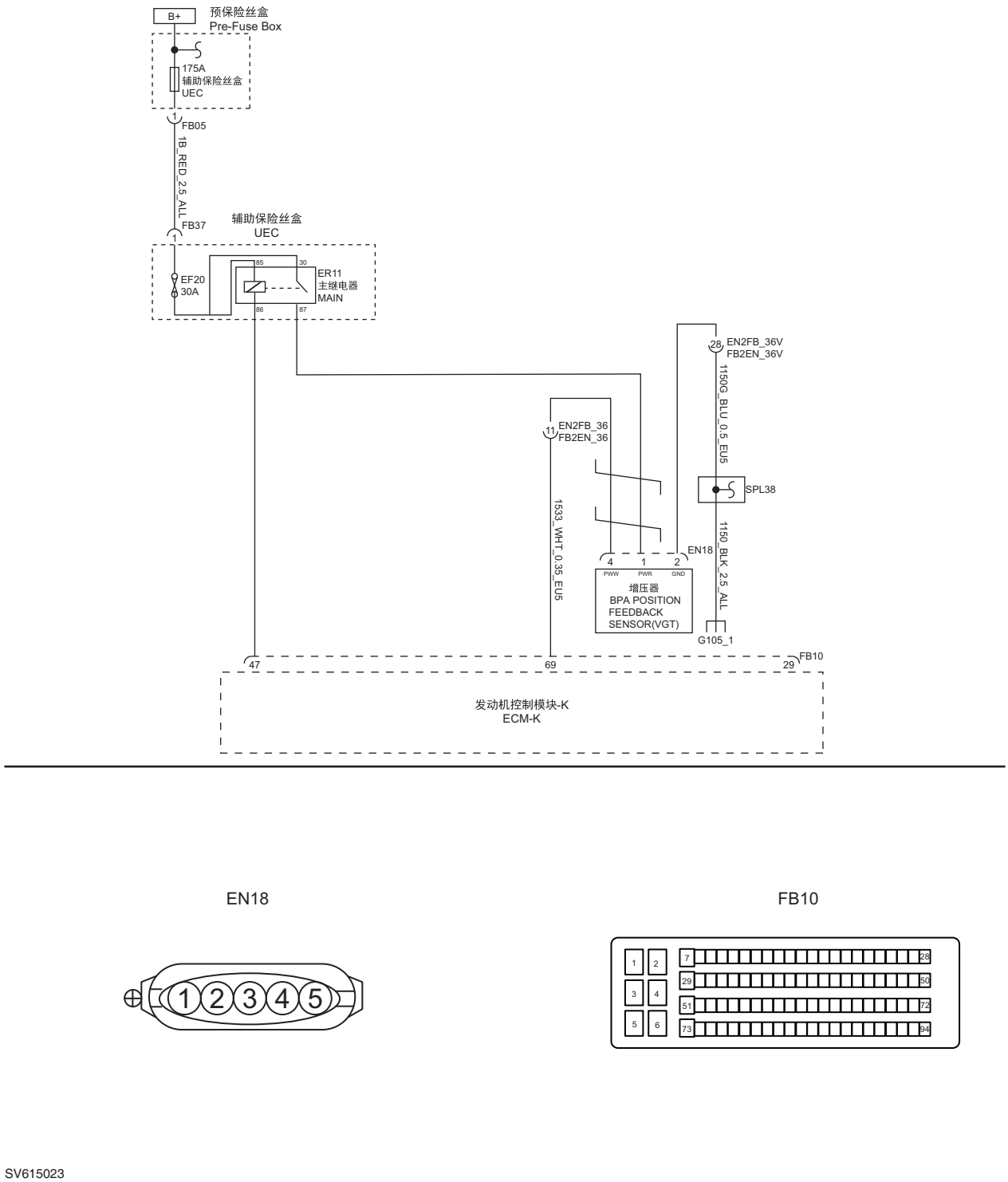
P0045、P0047、P0048

Description of fault diagnosis codes
P0045-13：No load error for turbo charger powerstage
P0045-4B：Over temperature error for turbo charger powerstage
P0047-11：Short circuit to ground error of turbo charger powerstage
P0048-12：Short circuit to battery error of turbo charger powerstage
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Pressurizer fault • Engine control module fault.
Description of circuit diagram
The supercharger is connected to the engine control module via wire, the main relay provides 12v voltage to the supercharger, and the engine control module will control the supercharger operation according to the engine status.

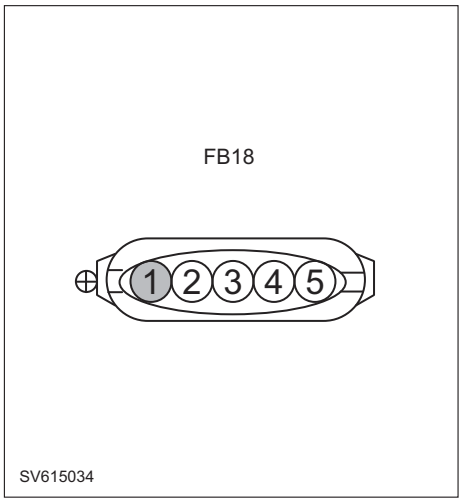
Circuit diagram

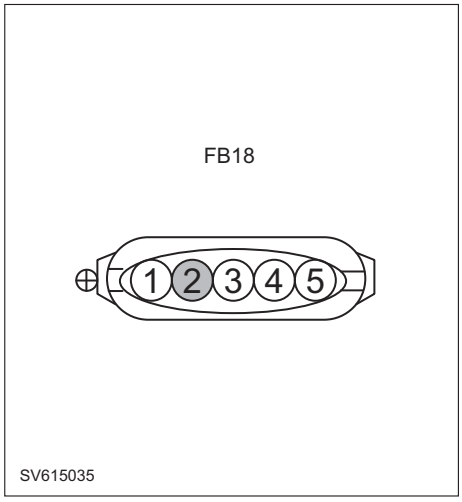
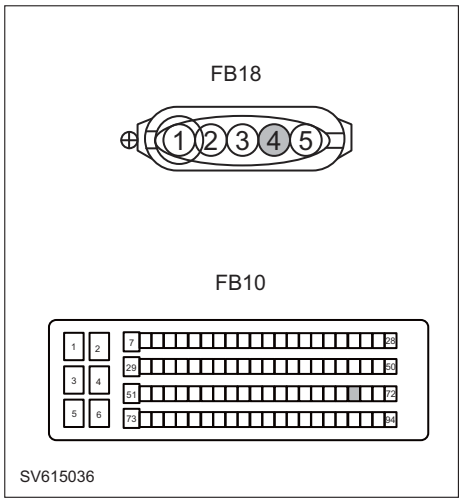


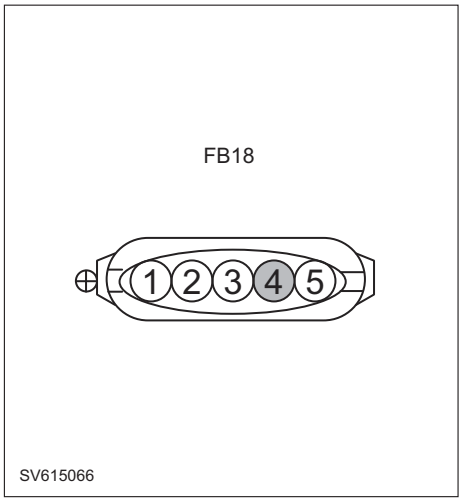
EN18

FB10

SV615023

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse EF20 for damage. 2 Check battery capacity. 3 Check the engine control module and pressurizer harness plug for damage, poor contact, aging and looseness. 4 Check whether the result is normal.
		No → Repair or replace the faulted parts.
Yes ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 Is there any other fault code except for P0045,P0047,P0048?
		Yes → Reference: DTC Summary List (EU5).
No ↓		
Step 3	Check the power supply circuit of pressurizer.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of pressurizer. 3 Turn the ignition switch to ON position. 4 Check the voltage between pressurizer harness plug and ground. Measure circuit: the voltage between EN18 plug terminal 1 and ground. **Standard value: 10~14V** 5 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		

Step 4	Check the ground circuit of pressurizer.
	1 Turn the ignition switch to OFF. 2 Check the resistance between pressurizer plug and ground. Measure circuit: the resistance between EN18 plug terminal 2 and ground. **Standard value: ≈ 0Ω** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the control circuit of pressurizer (open circuit).
	1 Disconnect the harness plug of engine control module. 2 Disconnect the harness plug of pressurizer. 3 Check the resistance of harness plug between pressurizer and engine control module. Measure circuit: the resistance between EN18 plug terminal 4 and FB10 plug terminal 69. **Standard value: ≈ 0Ω** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 6	Check the control circuit of pressurizer (short circuit) .
	1 Measure the resistance between the harness plug of rear oxygen sensor and ground. Measure circuit: the resistance between EN18 plug terminal 4 and ground. **Standard value: ∞** 2 Measure the voltage between the harness plug of rear oxygen sensor and ground. Measure circuit: the voltage between EN18 plug terminal 4 and ground. **Standard value: 0V** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 7	Check the control signal of pressurizer.
	1 Connect the harness plug of engine control module. 2 Turn the ignition switch to ON position. 3 Measure the voltage between the harness plug of pressurizer and ground. Measure circuit: the voltage between EN18 plug terminal 4 and ground. **Standard value: 0.05~0.2V** 4 Check whether the result is normal?
	No → Replace engine control module.
Next ↓	
Step 8	Complete the check and confirm whether the system works normally nor not.